

Sahel Torkamani

Research Interests

Theoretical Machine Learning, Differential Privacy, Zero-Knowledge Proofs.

Education

2024– **University of Edinburgh,**

Ongoing *PhD in Computer Science, Supervised by Professor Rik Sarkar and Professor Henry Gouk.*

2019–2023 **Sharif University of Technology,**

Bachelor of Applied Mathematics.

2013–2019 **Farzanegan 1 High-School,**

Diploma in Physics and Mathematics Discipline,

National Organization for the Development of Exceptional Talent.

Publications and Preprints

Under Review **Generalization in Neural Networks Through the Lens of Magnitude Potential,**
Sahel Torkamani, Henry Gouk, Rik Sarkar.

Under Review **Privacy Requires Slicing: Differentially Private Magnitude via Projections,**
Sahel Torkamani, Henry Gouk, Rik Sarkar.

ICML 2026 **Magnitude Distance: A Geometric Measure of Dataset Similarity ([arxiv](#)),**
Sahel Torkamani, Henry Gouk, Rik Sarkar.

AsiaCCS 2026 **PrivaDE: Privacy-preserving Data Evaluation for Blockchain-based Data Marketplaces ([arxiv](#)),**
Thomas Wong, Sahel Torkamani, Michele Ciampi, Rik Sarkar.

FORC 2025 **Count on your elders: Laplace vs Gaussian noise ([arxiv](#)),**
Joel Daniel Andersson, Rasmus Pagh, Teresa Anna Steiner, Sahel Torkamani.

TPDP 2024 **Improved Counting under Continual Observation with Pure Differential Privacy ([arxiv](#)),**
Joel Daniel Andersson, Rasmus Pagh, Sahel Torkamani.

JSAIT 2023 **Optimal Differential Privacy via Graphs ([arxiv](#)),**
Sahel Torkamani, Javad B. Ebrahimi, Parastoo Sadeghi, Rafael G. L. D'Oliveira, Muriel Médard.

ISIT 2022 **Heterogeneous Differential Privacy via Graphs ([arxiv](#)),**
Sahel Torkamani, Javad B. Ebrahimi, Parastoo Sadeghi, Rafael G. L. D'Oliveira, Muriel Médard.

Honor and Awards

2026 **Research Spotlight Speaker Prize at London Hopper Colloquium 2026,** *University College London.*

2026 **Awarded one-year BCS membership,** *British Computer Society.*

2024 **Awarded scholarship from the Informatics Graduate School,** *University of Edinburgh.*

2024 **Awarded for the best speaker in undergraduate research seminars on graph theory and computer science,** *Sharif University of Technology.*

2019 **Awarded scholarship from the National Elites Foundation.**

2018 **National Silver Medal,** *Iranian Mathematics Olympiad.*

2017 **National and International Gold Ruler,** *Iranian Geometry Olympiad, Advance Level.*

2017 **National Silver Medal,** *Iranian Mathematics Olympiad.*

2016 **National Silver Ruler,** *Iranian Geometry Olympiad, Medium Level.*

2015 **National Gold and International Bronze Ruler,** *Iranian Geometry Olympiad, Elementary Level.*

Internships and Summer Schools

- 2022 **Research Internship on Coding Theory and Machine Learning**,
Supervised by Professor Marco Mondelli, Institute of Science and Technology Austria,
Designed a sparse multi-decoder via recursive projection aggregation for Reed-Muller codes, implementing selection heuristics with a neural network.
- 2016 & 2017 **Iran's National Olympiad of Mathematics Summer Camp**,
For elite students winning a 3 round national competition.

Talks and Presentations

- Spring 2026 **Tracking Geometry of Data Representation to Explain Learning**,
RESEARCH SPOTLIGHT FINALIST AT LONDON HOPPER COLLOQUIUM 2026.
- Winter 2025 **Magnitude Distance: A Geometric Measure of Dataset Similarity**,
WORKSHOP ON PRINCIPLES OF GENERATIVE MODELING (PRIGM) @ EURIPS2025.
- Summer 2025 **Metric Space Magnitude for Machine Learning**,
INSTITUTE OF SCIENCE AND TECHNOLOGY AUSTRIA (ISTA).
- Summer 2022 **Heterogeneous differential privacy via graphs**,
IEEE INTERNATIONAL SYMPOSIUM ON INFORMATION THEORY (ISIT 2022).
- Summer 2022 **Optimal Binary Differential Privacy via Graphs**,
INSTITUTE OF SCIENCE AND TECHNOLOGY AUSTRIA (ISTA).
- Autumn 2021 **Introduction to Coding Theory, Mini-Course (video)**,
DEPARTMENT OF MATHEMATICAL SCIENCES, SHARIF UNIVERSITY OF TECHNOLOGY.

Teaching Experience

- Spring 2026 **Advanced Topics in Machine Learning**, *Teaching Assistant and Tutorial Lecturer*,
THE SCHOOL OF INFORMATICS, UNIVERSITY OF EDINBURGH.
- Autumn 2025 **Probabilistic Modelling and Reasoning**, *Teaching Assistant and Tutorial Lecturer*,
& 2026 THE SCHOOL OF INFORMATICS, UNIVERSITY OF EDINBURGH.
- Spring 2025 **Machine Learning Theory**, *Teaching Assistant and Tutorial Lecturer*,
THE SCHOOL OF INFORMATICS, UNIVERSITY OF EDINBURGH.
- Autumn 2024 **Discrete Mathematics and Probability**, *Tutorial Lecturer*,
& 2025 THE SCHOOL OF INFORMATICS, UNIVERSITY OF EDINBURGH.
- Autumn 2023 **Information Theory**, *Teaching Assistant (Unofficial)*,
DEPARTMENT OF MATHEMATICAL SCIENCES, SHARIF UNIVERSITY OF TECHNOLOGY.
- Autumn 2021 **Statistics and Applications**, *Teaching Assistant (Unofficial)*,
DEPARTMENT OF MATHEMATICAL SCIENCES, SHARIF UNIVERSITY OF TECHNOLOGY.
- Spring 2021 **Introduction to Algorithms and Python**, *Lecturer*,
YASAN ACADEMY.
- Spring 2020 **Algebra and Number Theory in Mathematics Olympiad**, *Lecturer*,
NATIONAL ORGANIZATION FOR DEVELOPMENT OF EXCEPTIONAL TALENTS, FARZANEGAN 1.

Selected Reviewing

- 2024 IEEE International Symposium on Information Theory (ISIT)

Skills

Programming Python, Java, R, Matlab

Frameworks PyTorch, NumPy, CUDA, Jupyter, Conda, SLURM

Machine Learning Generative Modelling, Geometric Deep Learning, Memorization & Generalization in Neural Networks,
Kernel Methods & Metric Space Theory, Differential Privacy

Languages Persian (Native), English (IELTS Band Score 7, TOEFL 101), Italian (B1)